

11. class, Electrodynamics and Optics

Page 891, Task 1. Displacement Current and the General Form of Ampère's Law

A 0.200-A current is charging a capacitor that has circular plates 10.0 cm in radius. If the plate separation is 4.00 mm,

- (a) what is the time rate of increase of electric field between the plates?
- (b) What is the magnetic field between the plates 5.00 cm from the center?

Page 891, Task 2. Maxwell's Equations and Hertz's Discoveries

A very long, thin rod carries electric charge with the linear density 35.0 nC/m. It lies along the x axis and moves in the x direction at a speed of 1.50×10^7 m/s.

- (a) Find the electric field the rod creates at the point ($x = 0$, $y = 20.0$ cm, $z = 0$).
- (b) Find the magnetic field it creates at the same point.
- (c) Find the force exerted on an electron at this point, moving with a velocity of $(2.40 \times 10^8) \hat{i}$ m/s.

Not from the Serway book:

Given a very long coaxial pair of ideal conducting cylinders with radii r_1 and r_2 . A voltage ΔU is applied between the outer and inner cylinders at one end of the pair, and at the other end a load resistor of resistance R is connected between the outer and inner cylinders. The coaxial cylinder pair functions as a transmission line; in both cylinders currents of equal magnitude but opposite direction flow, parallel to the axis.

- (a) Determine the electric field in the region between the cylinders.
- (b) Determine the magnetic field in the region between the cylinders.
- (c) Determine the Poynting vector field in the region between the cylinders.
- (d) Show that the surface integral of the Poynting vector field over a plane perpendicular to the cylinder axis is equal to the power delivered to the load.

And:

In a solenoid (with given length L , cross-section $A = \pi r^2$, and number of turns N), the current in the coil is increased uniformly in time.

- (a) Determine the magnetic field inside the coil as a function of time.
- (b) Determine the induced eddy (circulating) electric field inside the coil as a function of time.
- (c) Determine the Poynting vector field inside the coil.
- (d) Compare the time derivative of the energy stored in the coil with the surface integral of the Poynting vector taken over the lateral surface of the coil.

Page 892, Task 10.

Verify by substitution that the following equations are solutions to Equations 33.19 and 33.20, respectively:

$$E = E_{\max} \cos(kx - \omega t)$$

$$B = B_{\max} \cos(kx - \omega t)$$

$$\frac{\partial^2 E}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 E}{\partial t^2} \quad (33.19)$$

$$\frac{\partial^2 B}{\partial x^2} = \mu_0 \epsilon_0 \frac{\partial^2 B}{\partial t^2} \quad (33.20)$$

Page 892, Task 18.

Assuming the antenna of a 10.0-kW radio station radiates spherical electromagnetic waves, (a) compute the maximum value of the magnetic field 5.00 km from the antenna and

(b) state how this value compares with the surface magnetic field of the Earth.

Page 893, Task 20.

At one location on the Earth, the rms value of the magnetic field caused by solar radiation is 1.80 μT . From this value, calculate

(a) the rms electric field due to solar radiation,

(b) the average energy density of the solar component of electromagnetic radiation at this location, and

(c) the average magnitude of the Poynting vector for the Sun's radiation.

Page 893, Task 21.

A 25.0-mW laser beam of diameter 2.00 mm is reflected at normal incidence by a perfectly reflecting mirror. Calculate the radiation pressure on the mirror.